

Divya Teja Reddy Chekkera

Technical Support Engineer

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PROFILE

Technical Support Engineer – Verification with 1+ year of hands-on experience in ASIC/IP verification using SystemVerilog, UVM, SVA, and RAL-based register verification at Maven Silicon, Bengaluru. Skilled in building modular UVM testbench environments for AMBA protocol verification (AXI, AHB, APB, SPI, UART) and performing formal property verification using Cadence JasperGold — writing SVA properties, identifying RTL bugs through vacuous proofs, and achieving exhaustive corner-case coverage beyond simulation. Proficient in VCS, Questa Sim, Xcelium, and Verdi for simulation, debug, and coverage closure.

EXPERIENCE

Technical Support Engineer – Verification

Dec 2025 – Present

Maven Silicon Softech Pvt. Ltd. · Bengaluru, India

- Developed UVM-based testbench for APB-based SPI Controller — modular components including APB master agent, SPI slave agent, virtual sequencer, self-checking scoreboard, and functional coverage collector.
- Implemented RAL model for SPI register-level verification — front-door (APB transactions) and back-door (direct DUT) access, covering reset values, R/W permissions, and reserved bit behavior across all configuration registers.
- Performed formal property verification using Cadence JasperGold FPV — developed and proved SVA properties for register read/write behavior, APB protocol compliance, and reset correctness, achieving exhaustive proof coverage beyond constrained-random simulation.
- Identified and root-caused 5 RTL bugs surfaced through vacuous proofs and unreachable cover properties during formal verification; documented findings in a formal verification report.
- Achieved 100% functional coverage across SPI verification environments using constrained-random test sequences targeting all 4 SPI modes (CPOL/CPHA), FIFO boundary conditions, and error injection scenarios.
- Executed regression test suites using VCS, Questa Sim, and Xcelium — analyzed coverage reports and drove verification closure using Verdi waveform debug flow.
- Implemented SystemVerilog Assertions (SVA) to validate protocol-level behavior across APB and SPI interfaces, catching protocol violations early in the verification cycle.

Design and Verification Trainee

Nov 2024 – Nov 2025

Maven Silicon Softech Pvt. Ltd. · Bengaluru, India

- Built complete UVM testbench for AHB-to-APB Bridge verification — achieved 100% functional coverage and reduced bug detection time by 25%; components: agents, monitors, drivers, and scoreboard using VCS and Questa Sim.
- Verified APB-based UART 16550 controller — UVM testbench with APB and UART agents, RAL model for UART configuration registers (baud rate divisor, line control, FIFO control, interrupt enable), and SVA assertions for frame integrity.
- Achieved 100% functional coverage across UART operating modes: baud rates, data widths (5–8 bits), parity modes (odd/even/none), stop bit configurations, using constrained-random sequences.
- Designed a 1x3 packet router in SystemVerilog with FIFO buffering and arbitration logic — reduced packet latency by 30%, passing all corner-case verification tests.
- Completed 900 hours of industry-level VLSI training covering RTL design, UVM methodology, AMBA protocol verification, and formal verification techniques.

Cybersecurity Intern

Apr 2024 – Jun 2024

Tata Consultancy Services (TCS) · Hyderabad, India

- Conducted vulnerability assessments and penetration testing using Splunk SIEM for log analysis and security monitoring.
- Worked on incident response and log-parsing scripting, developing automation skills applicable to verification regression and debug workflows.

TECHNICAL SKILLS

Languages: SystemVerilog, Verilog, SVA (SystemVerilog Assertions), Python

Verification: UVM, RAL (Register Abstraction Layer), Constrained-Random Verification, Functional Coverage, Code Coverage, Assertion-Based Verification (ABV), Formal Verification, Directed Testing, Regression Analysis

EDA Tools: Synopsys VCS, Mentor Questa Sim, Cadence Xcelium, Cadence JasperGold (FPV), Synopsys Design Compiler

Protocols: AMBA (AHB, APB, AXI), AHB-to-APB Bridge, APB-SPI, APB-UART 16550

UVM Components: Agent, Driver, Monitor, Sequencer, Virtual Sequencer, Scoreboard, Environment, Coverage Collector

KEY PROJECTS

Formal Property Verification — APB-based SPI Controller (JasperGold FPV) · Formal Property Verification — APB-based UART Protocol (JasperGold FPV) · APB-based UART 16550 Functional Verification (UVM, RAL) · APB-based SPI Protocol Verification (UVM, RAL) · AHB-to-APB Bridge Verification (UVM) · 1x3 Packet Router RTL Design and Verification

ACHIEVEMENTS

- Performed formal verification using Cadence JasperGold — identified 5 RTL bugs through vacuous proofs and unreachable cover properties in APB-SPI bridge, achieving proof coverage not possible through simulation alone.
- Achieved 100% functional coverage across 3 verification environments (AHB-APB Bridge, APB-SPI, APB-UART 16550) — covering all protocol corner cases, error injection scenarios, and register field combinations.
- Built RAL models for SPI and UART controllers enabling front-door and back-door register verification — covering reset values, R/W permissions, and reserved bit behavior across all configuration registers.
- Published research paper in JETIR (March 2024) on Comparative Analysis of 32-Bit Complex Floating Point Multiplier.

EDUCATION

B.Tech — Electronics and Communication Engineering

R.V.R & J.C College of Engineering, Guntur · CGPA: 7.37 · 2020–2024

Intermediate (10+2)

Sri Chaitanya Jr College, Guntur · CGPA: 8.6 · 2018–2020

CERTIFICATIONS

- Advanced VLSI Design and Verification · Maven Silicon Softech Pvt. Ltd. · Jul 2025 · 900 hours · Grade B · ID: MS/23-24/3058
- Machine Learning · Internshala · 2023
- VLSI Design · Udemy

PUBLICATION

- Comparative Analysis of 32-Bit Complex Floating Point Multiplier · Journal of Emerging Technologies and Innovative Research (JETIR) · March 2024 · jetir.org/view?paper=JETIR2404815